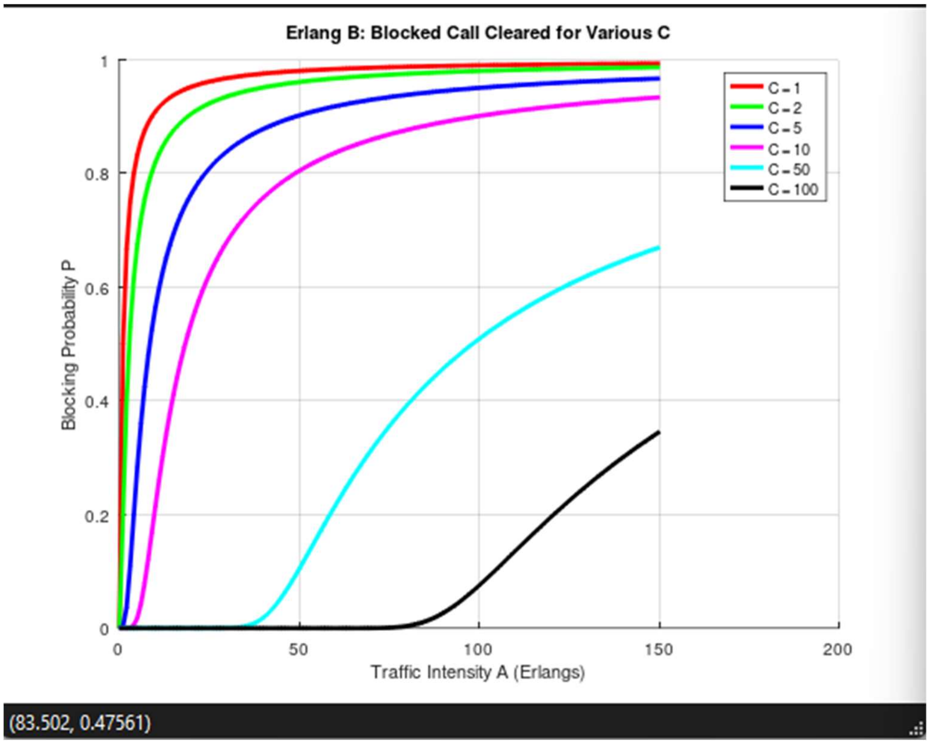
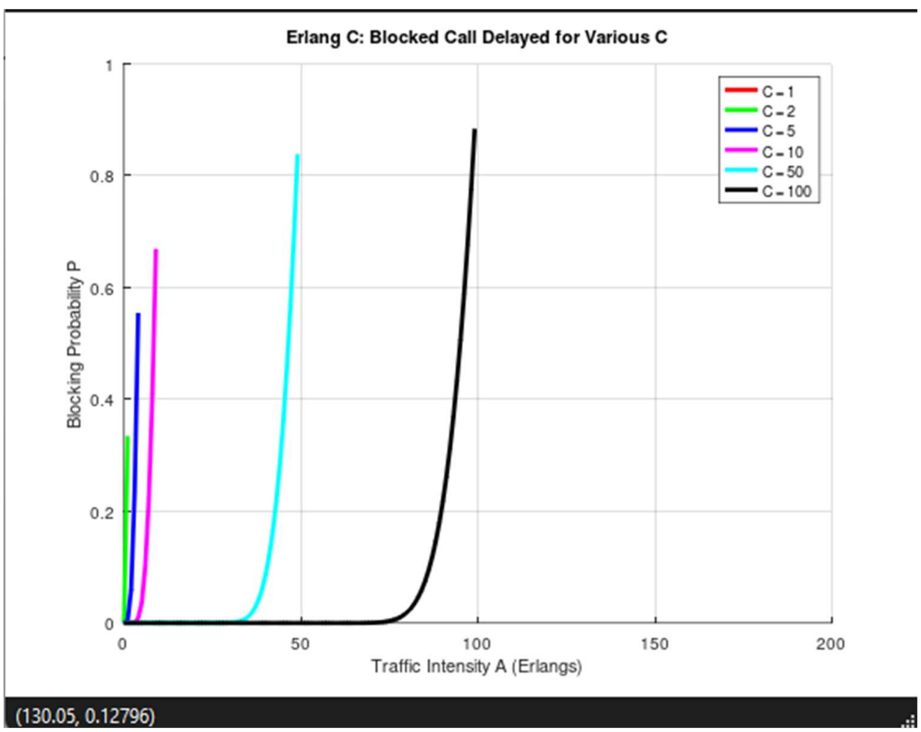
```
clc; clear all;

C_values = [1, 2, 5, 10, 50, 100];
A = 0:1:150;
colors = ['r','g','b','m','c','k'];

% ----- ERLANG B -----
figure;
hold on;

for idx = 1:length(C_values)
  C = C_values(idx);
  P = zeros(size(A));

  for k = 1:length(A)
    numerator = (A(k)^C)/factorial(C);
    denominator = sum(A(k).^(0:C) ./ factorial(0:C));
    P(k) = numerator / denominator;
  end

  plot(A, P, [colors(idx) '-'], 'DisplayName', ['C = ' num2str(C)], 'LineWidth', 2);
end

xlabel('Traffic Intensity A (Erlangs)');
ylabel('Blocking Probability P');
title('Erlang B: Blocked Call Cleared for Various C');
legend show;
grid on;
```

```
% ----- ERLANG C -----
figure;
hold on;

for idx = 1:length(C_values)
  C = C_values(idx);
  P = zeros(size(A));

  for k = 1:length(A)
    if A(k) < C
      sumC = sum(A(k).^(0:C-1) ./ factorial(0:C-1));
      numerator = (A(k)^C / factorial(C)) * (C / (C - A(k)));
      denominator = sumC + numerator;
      P(k) = numerator / denominator;
    else
      P(k) = NaN;
    end
  end

  plot(A, P, [colors(idx) '-'], 'DisplayName', ['C = ' num2str(C)], 'LineWidth', 2);
end

xlabel('Traffic Intensity A (Erlangs)');
ylabel('Blocking Probability P');
title('Erlang C: Blocked Call Delayed for Various C');
legend show;
grid on;
```


Erlang B: Blocked Call Cleared for Various C
Blocking Probability P
Traffic Intensity A (Erlangs)
C = 1
C = 2
C = 5
C = 10
C = 50
C = 100
(83.502, 0.47561)


Erlang C: Blocked Call Delayed for Various C
Blocking Probability P
Traffic Intensity A (Erlangs)
C = 1
C = 2
C = 5
C = 10
C = 50
C = 100
(130.05, 0.12796)